Undergraduate course documentation for ABET review

To: all instructors responsible for undergraduate courses – Spring 2001

From: Academic planning and program review committee
Steve Long, chair

The department must begin documenting our undergraduate courses in preparation for the ABET visit in Fall 2002. Every undergraduate course needs to be included in the self-study documentation. As most of you know, we now need to demonstrate that our course goals are addressing the ECE program outcomes and to the extent that is feasible, the design (professional) component that ABET expects to see.

We are trying to make the process of documenting your course as simple as we can for you while still providing us with material that can be directly included in the self-study report that must be compiled during the next academic year.

So, I'm asking you to please fill out the two documents below: a course outcome worksheet and a course syllabus. The content of these forms has been tailored to the ABET expectations.

1. Course Outcome Worksheet

The purpose for the document is to define the goals for your course and to demonstrate why you believe your course is achieving these goals. Also, it enables you to map your course goals to our department program outcomes and ABET's professional component. You may be interviewed during the ABET visit next year and asked to prove that you are adequately measuring the outcome of your course. Hopefully this form will help you build your case for the department.

You do not need to collect the samples of student work or statistics on grading this quarter but it will be necessary next spring. At that time, you should sample each entry on your table. You can use this form as a reminder next spring of what should be collected for ABET.

2. Course Syllabus

The syllabus is an image of the course as it stands at this time. It also includes a brief matrix summarizing the relationship between your course goals and the department program outcomes.

And for your reference: I have prepared samples for one of my courses – shown below. Also:

3. Program outcomes. These are the short term objectives (labeled P1 to P6) of the ECE department undergraduate instruction as discussed and modified following the 5/16/2001 faculty meeting.

4. Professional Component. This is from the ABET Criterion 2000 document. Not all courses will include this component, but indicate the extent to which your course does contribute.

Please return the forms to me by June 18, 2001

Course outcome worksheet (sample)

Course: ECE145A Communication Electronics

Instructor: Prof. Steve Long

Course Goals

1. Transmission Lines and Lumped Components
    1. Use S parameters and the Smith Chart for design of lumped element and distributed L matching networks
    2. Able to model (Agilent ADS) and measure (network analyzer) nonideal lumped components at high frequencies

2. Wideband and Bandpass Amplifiers:
    1. Able to use network analyzer to measure gain and phase of amplifiers
    2. Use Mason's gain rule to derive gain, reflection, and transmission coefficients.
    3. Able to analyze, design and build a wideband resistive feedback amplifier.
    4. Can use S-parameters with gain and stability circles to design narrowband amplifiers.
    5. Able to design stable DC bias circuits for amplifiers

3. Low noise design:
    1. Familiar with the use of signal to noise ratio, noise figure, noise temperature, and the measurement and simulation of noise figure.
    2. Can use two-port noise parameters, noise and available gain circles to analyze, design and build a low noise amplifier

Contribution to program outcomes and professional component.

Course Goals	**Course activity**	**Mapping to ECE Program Outcomes**	**Contribution to professional component** (design experiences that incorporate standards and constraints and build upon previous work)	**Material to be collected to verify outcome or goal**
1.1	Homework, Midterm exam	P2: In-depth training in specialty areas P4: design experiences	Design matching networks to meet given specifications.	Statistics on individual exam problems measuring comprehension and mastery of topics
1.2	Lab assignment #1. Use of network analyzer to evaluate components. Use of ADS simulation software to verify models.	P3: laboratory experiences		Samples of lab reports; grades on reports
2.1	Lab assignment #2. Use of network analyzer to evaluate amplifier.	P2: In-depth training; P3: laboratory experiences		Samples of reports; grades on reports

2.2-2.5	Lab assignment #2 Use of ADS simulation software to verify design of amplifier and compare with measurements Homework, final exam,	P2: In-depth training; P3: laboratory experiences P4: design experiences	Design, construct and evaluate amplifier to meet given specifications. Employ tradeoffs and constraints in the design. Consider cost and power requirements. Build upon ECE137B: feedback and biasing.	Statistics on homework and individual exam problems Samples of reports; grades on reports
2.4, 2.5, 3.1, 3.2	Lab assignment #3. Design and build a narrowband LNA. Use of noise figure meter. Use noise and stability analysis in ADS to design amplifier.	P2: In-depth training; P3: laboratory experiences P4: design experiences	Design, construct and evaluate amplifier to meet given specifications. Employ tradeoffs and constraints in the design. Consider cost and power requirements. Build upon goals 1 and 2; ECE137B.	Sample lab reports
3.1, 3.2	Homework final exam,	P2: In-depth training; P4: design experiences	Design amplifier to meet given specifications.	Statistics on homework and individual exam problems
1.2, 2.1, 3.1, 3.2	Lab projects 1 – 3	P5: team experiences and written communication skills		Sample lab reports; questions on team experience.

Prepared by: Steve Long **Date: June 6, 2001**

Course Syllabus (sample)

ECE145A Communication Electronics 5 units

Catalog description:

Analog communication circuits 1 MHz to 1 GHz with emphasis on receivers. S parameter design techniques, nonideal components, distortion, low noise amplifier design and characterization , system level analysis.

Prerequisites: ECE137AB

Text, References, and Software:

G. Gonzalez, *Microwave Transistor Amplifiers: Analysis and Design*, Second Ed., Prentice-Hall, 1997.
ECE145A lecture notes
Agilent Advanced Design System (ADS) simulation tools

Course Goals

1. Transmission Lines and Lumped Components
    1. Use S parameters and the Smith Chart for design of lumped element and distributed L matching networks
    2. Able to model (Agilent ADS) and measure (network analyzer) nonideal lumped components at high frequencies

2. Wideband and Bandpass Amplifiers:
    1. Able to use network analyzer to measure gain and phase of amplifiers
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    5. Able to design stable DC bias circuits for amplifiers

3. Low noise design:
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    2. Can use two-port noise parameters, noise and available gain circles to analyze, design and build a low noise amplifier

Class/Laboratory hours

Two 75 minute lectures per week. Laboratory is open by card key access. Minimum of 6 hours per week is required.

Contribution to program outcomes

Course Goals	P1	P2	P3	P4	P5	P6
1.1		X		X		
1.2			X		X	
2.1			X		X	
2.2		X		X		
2.3		X		X		
2.4		X		X		
2.5		X	X	X		
3.1			X		X	

3.2		X	X	X	X	

Professional component

Lab projects 2 and 3 require the design of amplifiers with given specifications, constraints (economic and environmental) and design tradeoffs. The material builds upon biasing, feedback analysis and bandwidth estimation techniques from ECE137B.

Prepared by: Steve Long **Date: June 6, 2001**

Program Outcomes

The ECE program expects our students upon graduation to have:

P1. Acquired strong basic knowledge and skills in those fundamental areas of mathematics, science, and electrical engineering that are required to support specialized professional training at the advanced level and to provide necessary breadth to the student's overall program of studies. This provides the basis for lifelong learning.

P2. Experienced in-depth training in state-of-the-art specialty areas in electrical engineering. This is implemented through our senior electives. Students are required to take two sequences of at least two courses each at the senior level.

P3. Benefited from imaginative and highly supportive laboratory experiences where appropriate throughout the program. The laboratory experience will be closely integrated with coursework and will make use of up-to-date instrumentation and computing facilities. Students should experience both hardware-oriented and simulation-oriented exercises.

P4. Experienced design-oriented challenges that exercise and integrate skills and knowledge acquired in several courses. These may include design of components or subsystems with performance specifications. Graduates should be able to demonstrate an ability to design and conduct experiments as well as analyze the results.

P5. Learned to function well in teams. Also, students must develop communication skills, written and oral, both through team and classroom experiences. Skills including written reports, web page preparation, and public presentations are required.

P6. Completed a well-rounded and balanced education through required studies in selected areas of fine arts, humanities, and social sciences. This provides for the ability to understand the impact of engineering solutions in a global and societal context. A course in engineering ethics is also required of all undergraduates.

The professional component will not be a part of all undergraduate courses, but please indicate the extent that your course does contribute to any of these requirements. Also, please consider whether you can add anything to your course that contributes to any of these goals. For example, considering: cost of the design (economic); power consumed (environmental); hazards in the clean room or in use of a design (environmental, health and safety)

Professional Component (From ABET Criteria 2000)

"Students must be prepared for engineering practice through the curriculum culminating in a major design experience based on the knowledge and skills acquired in earlier course work and incorporating engineering standards and realistic constraints that include most of the following considerations: economic; environmental; sustainability; manufacturability; ethical; health and safety; social; and political. The professional component must include:

(a) one year of a combination of college level mathematics and basic sciences (some with experimental experience) appropriate to the discipline

(b) one and one-half years of engineering topics, consisting of engineering sciences and engineering design appropriate to the student's field of study

(c) a general education component that complements the technical content of the curriculum and is consistent with the program and institutional objectives."